Two Knights

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1072 | Introductory Problems | https://cses.fi/problemset/task/1072

Your task is to count for $k=1,2,\ldots,n$ the number of ways two knights can be placed on a $k \times k$ chessboard so that they do not attack each other.

Input

The only input line contains an integer n.

Output

Print n integers: the results.

Constraints

- $1 \le n \le 10000$

Example

Input
8

Output
0 6 28 96 252 550 1056 1848